



STRIKING GENERICS: SHARKS ATTACK PEOPLE AND GLIPPETS LOVE TO PLAY WITH FIRE

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partially funded by the British Academy

Valence Asymmetries Seminar

20 January 2026

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A B C

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letters



finger
spaces



full
stop

Collaborative work



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Overview slide

- Generic Generalisations
- Striking generics
 - Sharks attack people
- Background on generics
 - a classification of generics
 - prevalence & exception tolerance
- Part 1: acquisition of strikingness & exception tolerance
- Part 2: comprehension & production of generics in individuals with autism
- Discussion and Conclusions

Generalisations

- Some cats like milk.
- Three cats like milk.
- A few cats like milk.
- Most cats like milk.
- All cats like milk.

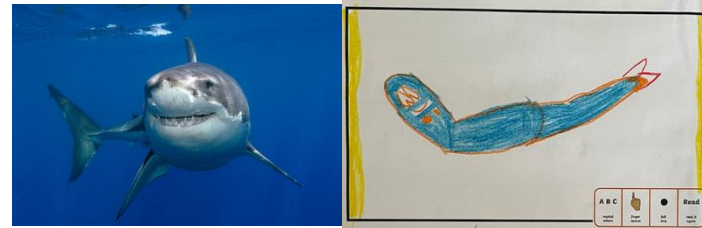


- **Cats** like milk.
- **A cat** likes milk.
- **The cat** likes milk.
- **Cats** drink milk after hunting.
- **Generics** are generally true yet allow for exceptions

Striking generics

- Low prevalence generics about a striking property
- “Property must only be exhibited by a small minority of the kind, and must signify something dangerous which is to be avoided” (Leslie 2007, 2008, Leslie et al. 2011)

- (1) Sharks attack people
- (2) Mosquitoes carry the West Nile virus
- (3) Lead toys poison children



- People accept *sharks attack people* as true even though a tiny minority of sharks do so (Prasada et al. 2013)
- NB: Sharks attack and kill 10 humans per year, on average. Humans, in contrast, annually kill about 20 to 30 million sharks according to the Florida Museum of Natural History's Department of Ichthyology

Leslie on striking generics

- “The sentence attributes harmful, dangerous, or appalling properties to the kind. More generally, if the property in question is the sort of property of which one would be well served to be forewarned, even if there were only a small chance of encountering it, then generic attributions of the property are intuitively true.” (Leslie 2008)
- “The disposition to generalize strikingly negative information on the basis of even a single event thus appears to be a pervasive aspect of our thinking.” (Leslie 2017)

A classification of generics

Prevalence and exception tolerance in generic statements

Cats are animals.

Quasi-definitional

Horses have four legs.

Majority characteristic

Ducks lay eggs.

Minority characteristic

Sharks attack people.

Striking

Cars have radios.

Majority statistical

Canadians are right-handed.

False Generalisation (majority)

Tigers are albino.

False Generalisation (minority)

(Leslie et al. 2011, Prasada et al. 2013)

Tolerance to exceptions

- Majority characteristic

Horses have four legs



Tolerance to exceptions

Minority characteristic

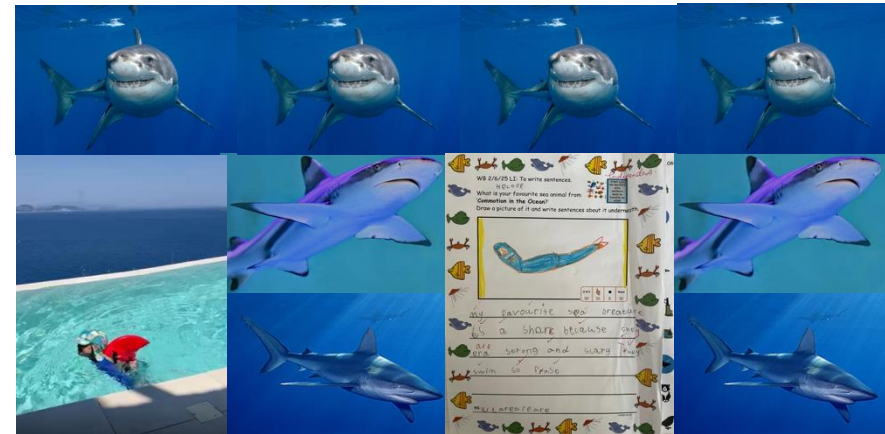
Ducks lay eggs



Tolerance to exceptions

Striking generics

Sharks attack people



Tolerance to exceptions

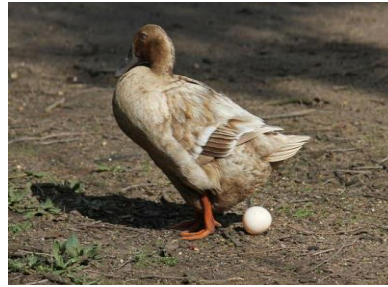
- Majority characteristic

Horses have four legs



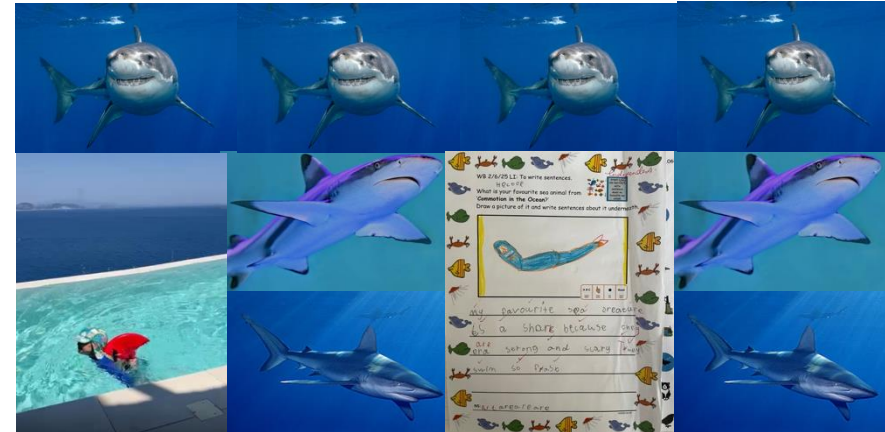
- Minority characteristic

Ducks lay eggs



- Striking generics

Sharks attack people



Exception tolerance and acceptance rates

- Neurotypical adults show exception tolerance to a wide range of generics and accept the various types of generics at different rates (e.g., Leslie et al. 2011; Prasada et al. 2013; Lazaridou-Chatzigoga et al. 2019, Castroviejo et al. 2023, Lazaridou-Chatzigoga et al. 2024)

Predication Type	Definition	Truth Value judgment (1-100%)	Prevalence estimates	Truth Value judgment (-3 to +3)	Truth Value judgment (1 to 7)
	Inferred prevalence	Leslie et al. 2011	Prasada et al. 2013	Prasada et al. 2013	Lazaridou-Chatzigoga et al. 2024
Quasi-definitional <i>Cats are animals</i>	100%	90%	86%	2.38	6.3
Majority Characteristic <i>Horses have four legs</i>	>50%	96%	90%	2.48	6.7
Minority Characteristic <i>Ducks lay eggs</i>	<50%	85%	59%	1.77	6
Majority (statistical) <i>Cars have radios</i>	>50%	82%	68%	1.40	5.6
Striking <i>Sharks attack people</i>	~5%	77%	33%	1.5	5.4
False generalisation Majority: <i>Canadians are right-handed</i> Minority: <i>Tigers are albino</i>	<50% >50%	no distinction 38%	64% 20%	.21 -.52	no distinction 3.7

Interim summary

- Striking generics have low prevalence (and very high exception tolerance)
- Adults accept them in truth value judgment tasks at rates that are lower than other types of generics, but still very high (~75%)

Issues arising

- **Acquisition**

- What do children know about striking generics and the relationship between strikingness and exception tolerance?

- **Neurodivergent populations**

- How do adults with autism understand and produce generics, and more particularly, striking generics?

Acquisition of strikingness and exception tolerance

Acquisition of generics

- Numerous studies by Gelman and colleagues argue that preschoolers understand generics and are able to distinguish them from non-generics (see Gelman, 2003, 2010, Leslie 2008; see Lazaridou-Chatzigoga et al., 2015 & Kirkpatrick 2024 for a critical review)
- Children seem to rely on generics when they learn about familiar and novel kinds and they use them in order to draw inferences (Hollander et al., 2002; Gelman and Bloom, 2007)
- Children seem to know that a property introduced with a generic is likely to be extended to new instances of the kind, both when they learn novel properties of familiar categories (Gelman et al., 2002; Chambers et al., 2008 with 4-year-olds) and when they learn properties of novel kinds (Graham et al., 2011 with 30-month-old children)

Child-friendly striking generics

- Child-friendly materials?
 - striking, noteworthy property
- For the preparation of the materials we used the Warriner et al. database (2013), who measured valence, arousal, and dominance ratings for 13,915 English lemmas on adult participants using a 9-point scale:
 - valence: the pleasantness of the stimulus
 - arousal: the intensity of emotion provoked by the stimulus
 - dominance: the degree of control exerted by the stimulus

examples at the extreme ends:

- lowest valence: pedophile 1.26 – highest valence: vacation 8.53
- lowest arousal: grain 1.6 – highest arousal: insanity 7.79
- lowest dominance: dementia 1.68 – highest dominance: paradise 7.9

Child-friendly striking generics

- We identified words that have high arousal, which we used as proxy for 'striking'
- In the final set of items used, striking properties had high arousal (range 4.91–7.24, mean 6.08) and average valence (range 2.53–4.68, mean 3.68), while neutral properties had average arousal (range 3.14–5.24, mean 3.99) and high valence (range 6.09–7.5, mean 6.85).
- We also cross-checked these against the Kuperman et al. (2012) database for AoA (Age of Acquisition) of 30,000 English words in order to make sure our participants would know the words we would use.

Study design

- Our child study had the following manipulations (between-subjects design):
 - a. nominal: generic vs. specific: *borps* vs. *these borps*
 - b. property: neutral vs. striking, e.g. *borps love to talk to their mothers* vs. *borps love to scare their mothers*
 - c. exceptions: minimal or maximal (1 or 3 exceptions presented)

Two studies

- *Exp 1*
- 32 English-speaking 4- and 5-year-olds, recruited from schools in London
- 140 English native adults, recruited via prolific academic
- *Exp 2*
- 32 English-speaking 4- and 5-year-olds, recruited from schools in London
- 152 English native adults, recruited via prolific academic

Materials



Neutral	Striking
Ackles love to play with toys.	Ackles love to play with fire.
Borps love to talk to their mothers.	Borps love to scare their mothers.
Glippets love to run through parks.	Glippets love to smash through walls.
Murbs love to draw their names.	Murbs love to shout their names.
Pagons love to feel safe.	Pagons love to feel afraid.
Scobbits love to make new games.	Scobbits love to cheat at games.
Vardies love to play with cats.	Vardies love to play with snakes.
Zorbs love to make people sing.	Zorbs love to make people angry.

Exp 1: Generic neutral/**striking** condition

- Step 1:



“These are ackles. Ackles love to play with toys/**fire**.”

- Step 2:



“Does this ackle love to play with toys/**fire**?”



Exp 2: Generic neutral/**striking** condition

- Step 1:



“These are ackles. Ackles love to play with toys/**fire**.”

- Step 2:



minimal: “But not this one. This ackle doesn’t love to play with toys/**fire**.”



maximal: “But not these ones. These ackles don’t love to play with toys/**fire**.”

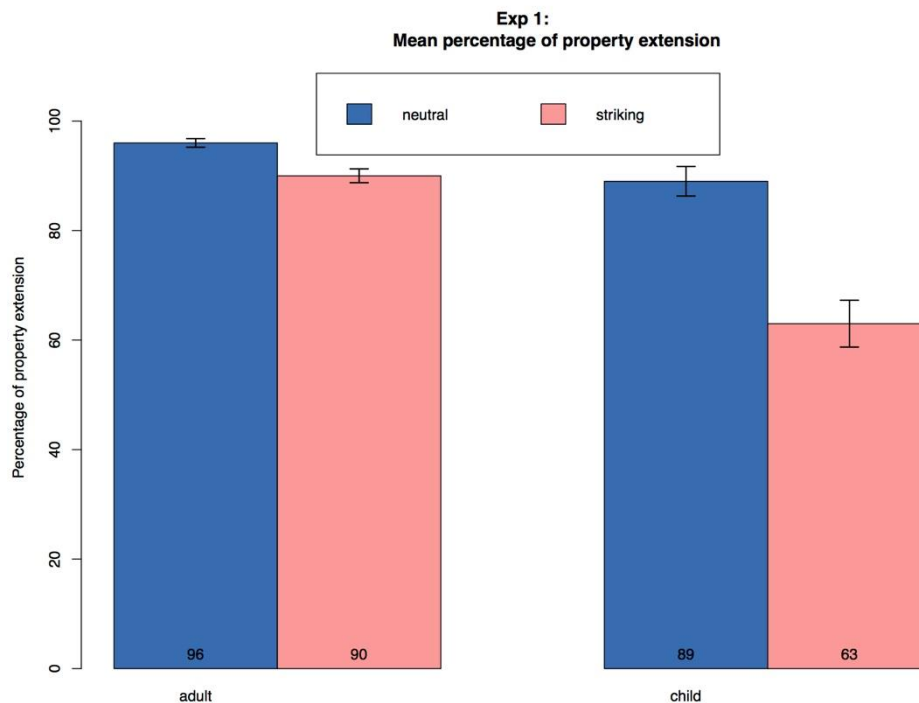
- Step 3:



“Does this ackle love to play with toys/**fire**?”



Experiment 1



R: glmer

main effect of property $p = .029$

main effect of age $p = .039$

no interaction

in both groups responses

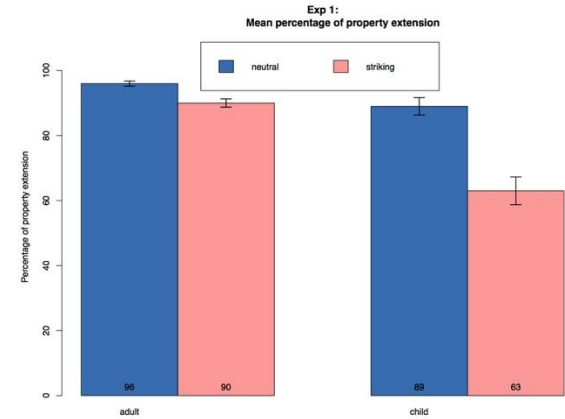
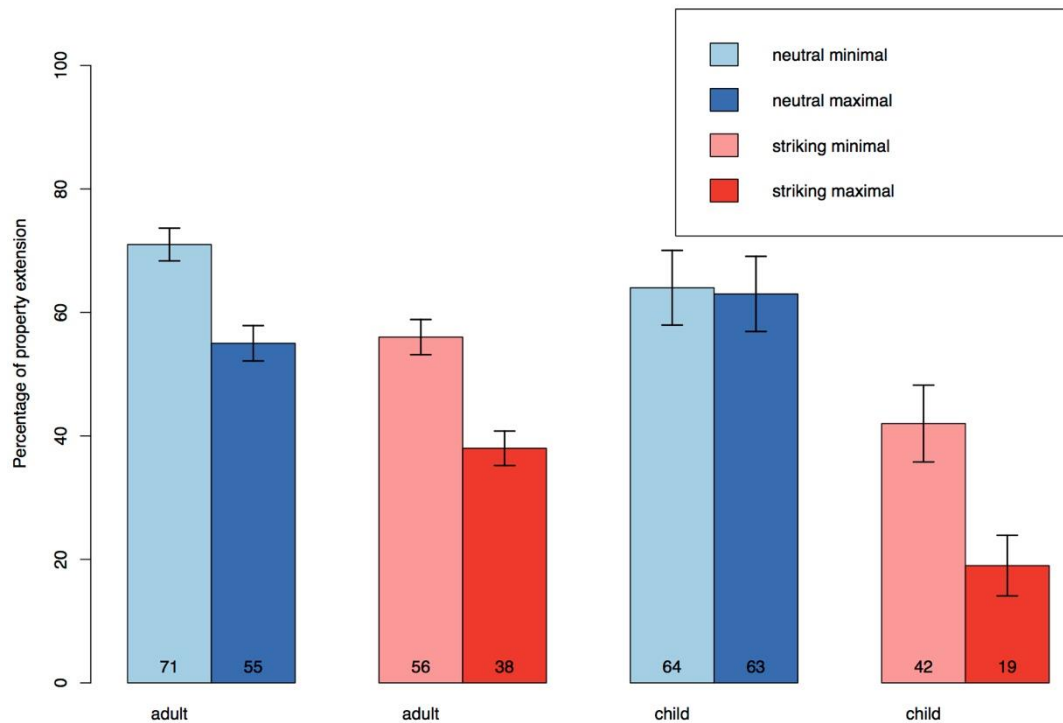
significantly different from chance

$ps < .001$

Lazaridou-Chatzigoga, Dimitra, Katsos, Napoleon, and Stockall, Linnaea. (2019) Generalizing About Striking Properties: Do Glippets Love to Play With Fire? *Frontiers in Psychology* 10:1971. doi: [10.3389/fpsyg.2019.01971](https://doi.org/10.3389/fpsyg.2019.01971)

Experiment 2

Exp 2:
Mean percentage of property extension



glmer in R

- main effect of property $p < .001$
- main effect of exception $p < .001$
- no main effect of age $p = .84$
- no interactions
- only adults' responses differ significantly from chance $p = .03$

Lazaridou-Chatzigoga, Dimitra, Katsos, Napoleon, and Stockall, Linnaea. (2019) Generalizing About Striking Properties: Do Glippets Love to Play With Fire? *Frontiers*

in
Psychology 10:1971. doi: [10.3389/fpsyg.2019.01971](https://doi.org/10.3389/fpsyg.2019.01971)

Discussion

- The findings underscore that 4- and 5-year-old children are
 - sensitive to strikingness and to exception tolerance for generalizations
 - developing toward an adult-like behavior with respect to the interplay between strikingness and exception tolerance when they learn about novel kinds

Discussion

- Two options
- Option (a)
 - it could be that the fact that the property is striking and less common leads participants to be conservative and extend striking properties less than neutral properties - in line with Prasada et al. (2013) findings
- Option (b)
 - it could be that both children and adults would extend the property even more in the case of striking properties than with neutral ones in order to maximize transfer of important information about a novel kind (in line with Leslie 2008, Cimpian et al. 2010)
- Our results are consistent with option (a), the “conservative” option

Discussion

- Why?
- Methodology
 - Our task involved measuring extension to a new exemplar of a novel kind while Cimpian et al. (2010)'s task measured whether a given statement about a novel kind was true or false. Our task was also more immersive given the more naturalistic environment we adopted
- Operationalisation of strikingness
 - In an attempt to construct child-friendly striking properties, strikingness was only indirect or implicit in our study

Follow up

- More work on the licensing conditions of strikingness for adults is needed given that our results diverged from other results in the literature
- Using versions of our materials that make explicit mention of danger/distinctiveness only with adults in the first instance would help determine whether the way we operationalized strikingness affected the results
- The subsequent challenge would be to prepare versions of these explicit materials without being too alarming in order to be able to use them to test young children

Follow up

- Focus on negative striking properties
- What about *positing* striking ones?
 - People helping save someone else's life or making large donations cases are not as straightforward and it is not easy to come up with examples (Leslie 2017)
- Would positive striking properties receive similar extension rates to negative striking properties?
- How essential is the dimension of danger or threat for striking properties?

Follow up

- Some items we have come up with:
- Borps draw dinosaurs (positive striking) / Borps draw circles (neutral)
- Borps fly to school (positive striking) / Borps walk to school (neutral)
- Borps serve pizza (positive striking) / Borps serve bread (neutral)

Generalisations in autism

Autism and Generalisations

- How autistic people understand or produce generics has not been empirically studied before, while only some aspects of quantifier meaning (e.g., scalar inferences) have received attention (e.g., Pijnacker et al. 2009; van Tiel & Kissine 2018)
- Generics are everyday language and help us communicate beyond immediate experiences
- They are vehicles of knowledge (*Tigers have stripes*, *Hector cycles to school*) but can also be used to express stereotypes (*Lawyers are greedy*)
- → Language/cognition interaction

Background

- Two related phenomena:
 - Impaired exception-handling in reasoning tasks (Pijnacker et al. 2009)
 - Residual difficulties in categorical induction (Naigles et al. 2013)

Hypothesis

- Autistic people
 - are more sensitive to **Precisification** (Klecha 2018), i.e., ways of making a statement precise
 - have different standards of precision (Lewis, 1979)
- Marked intolerance to exceptions (as in the social domain: Strang et al., 2017)

Nuanced Precisification hypothesis

- **H Nuanced Precisification.** Autistic individuals have a strong preference for a strict standard of precision only in certain types of generics, because these have a larger number of exceptions compared to other types, shown as ***higher rejections rates*** [comprehension study]/ ***production of fewer generic statements*** [production study]:

- (a) *minority characteristic statements (Ducks lay eggs)* - systematic and predictable exceptions are the majority (i.e., male, baby, infertile ducks)
- (b) *rare but striking statements (Sharks attack people)* – exceptions are the norm

Low production rates are also expected in:

- (a) *false generalisations (Canadians are right-handed/ Tigers are albino)* – the generic form sounds false or mistaken when the property is prevalent among members of the kind and there is a sufficiently salient alternative property

Comprehension study: Methods

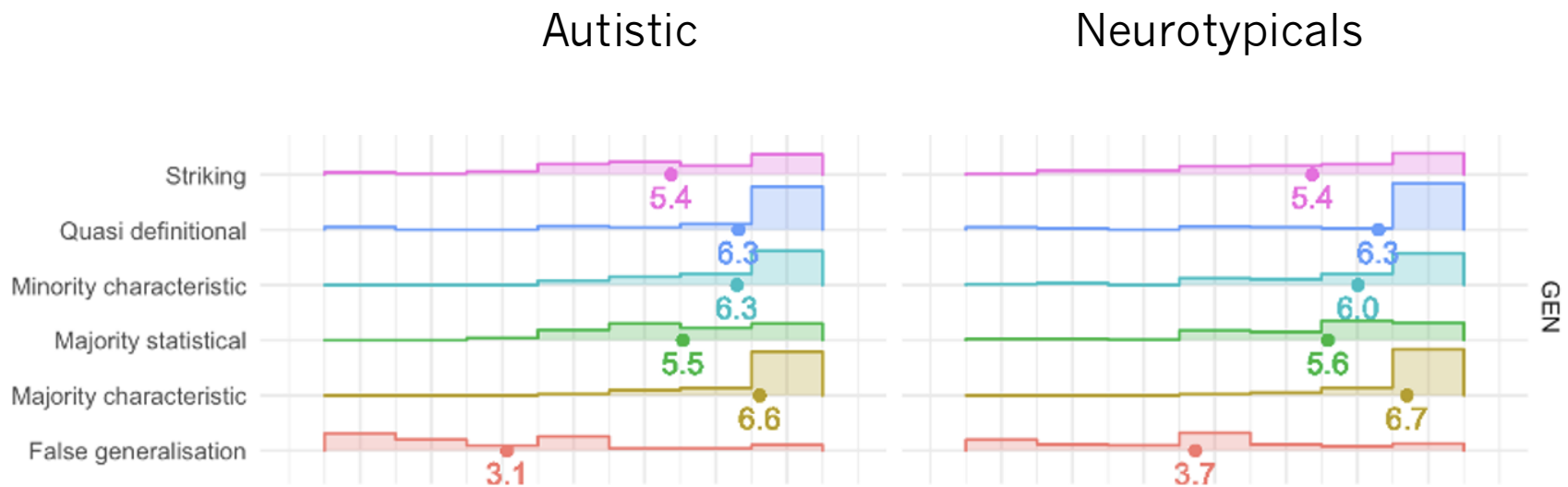
- 60 native English speakers (ongoing recruitment, Prolific)
 - 30 autistic, mean age: 35.4, 13 male, 16 female, 1 non-binary
 - 30 neurotypical, mean age: 32.9, 16 female, 14 male)
- 6 generic statement types
- 5 quantifier forms
 - {*All/Some/Most/No/∅* (GEN bare plurals)} *dogs have tails.*

<i>Statement type</i>	<i>Example</i>
Definitional	Cats are animals
Majority Characteristic	Horses have four legs
Minority Characteristic	Ducks lay eggs
Majority Statistical	Cars have radios
Striking	Sharks attack people
False generalisation	majority: Books are paperbacks minority: Tigers are albino

- Participants indicated agreement with a statement on a 1-7 rating scale (1: 'definitely not'; 7: 'definitely yes')
- 90 critical items; 90 fillers ('London is in England', 'Paris is in Japan')

Comprehension study

- Our comprehension study replicated the neurotypical findings and extended them to autistic adults



Group is not a significant predictor: No evidence of a difference

Production study: Methods

- 75 English native speakers (ongoing recruitment, Prolific)
 - 40 neurotypicals, mean age: 32.10, 22 female, 18 male
 - 35 individuals with autism, mean age: 32.15; 21 female, 2 non-binary, 12 male
- Autistic status (self-report and screening self-report questionnaire SRS-2):
 - 98.25 for neurodivergent, 45.96 for neurotypicals
- Verbal IQ (Mill Hill Synonyms task) and non-verbal IQ (Matrix Reasoning): within the typical range for both groups
- Education: Majority in both groups had at least BA degree

Production study: Task

- Teach a child things about our planet Earth and fill in the box with a word referring to quantity (e.g., *all* , *some*, etc.), avoiding the use of number words (3, 100)
- Leave the box blank if needed, i.e., leave the sentence as it is
- Be as precise and as general as necessary

(i) I would tell them that _____ cats are animals.
_____ horses have four legs.
_____ sharks attack people.

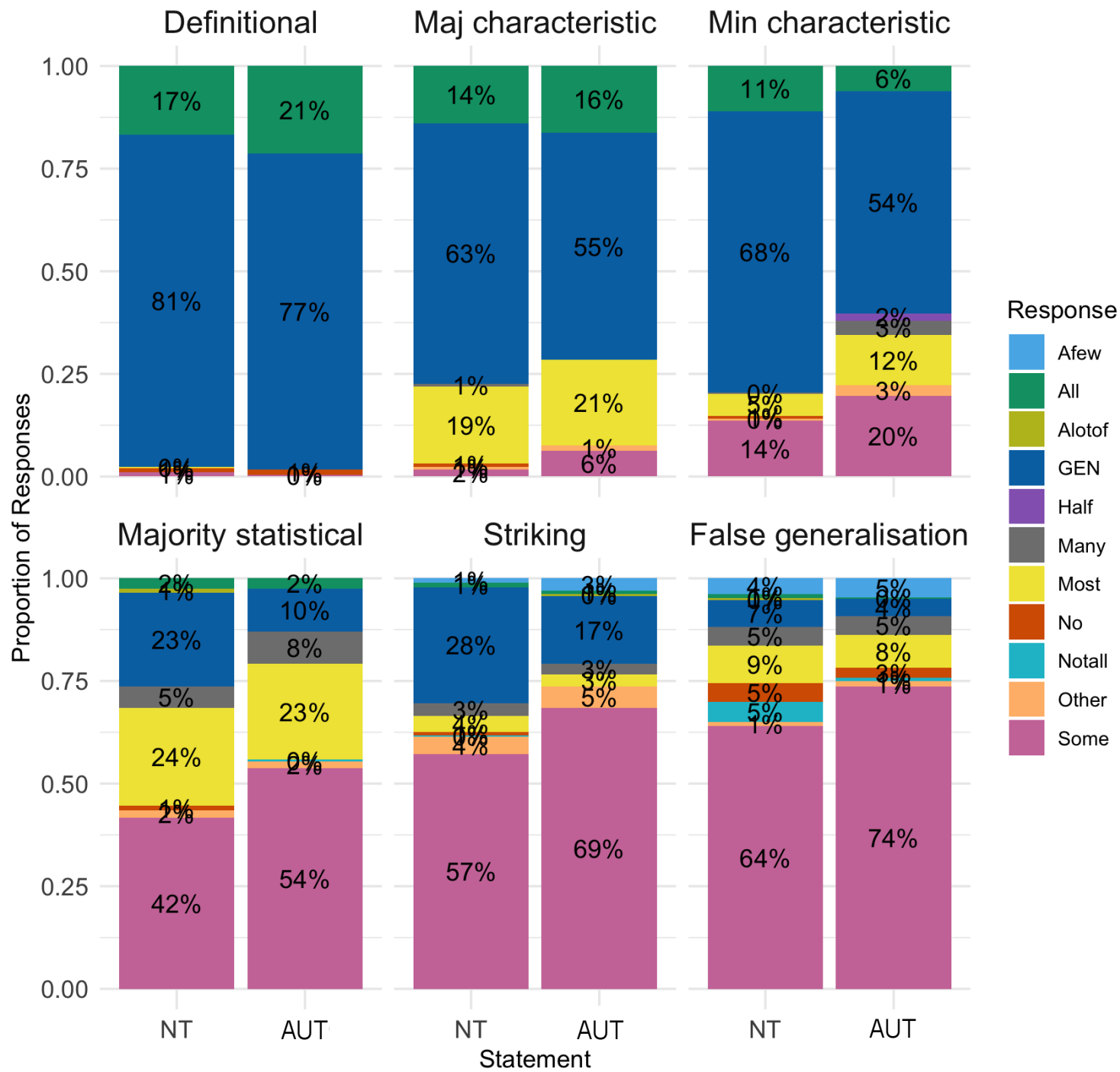
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Production study: Items

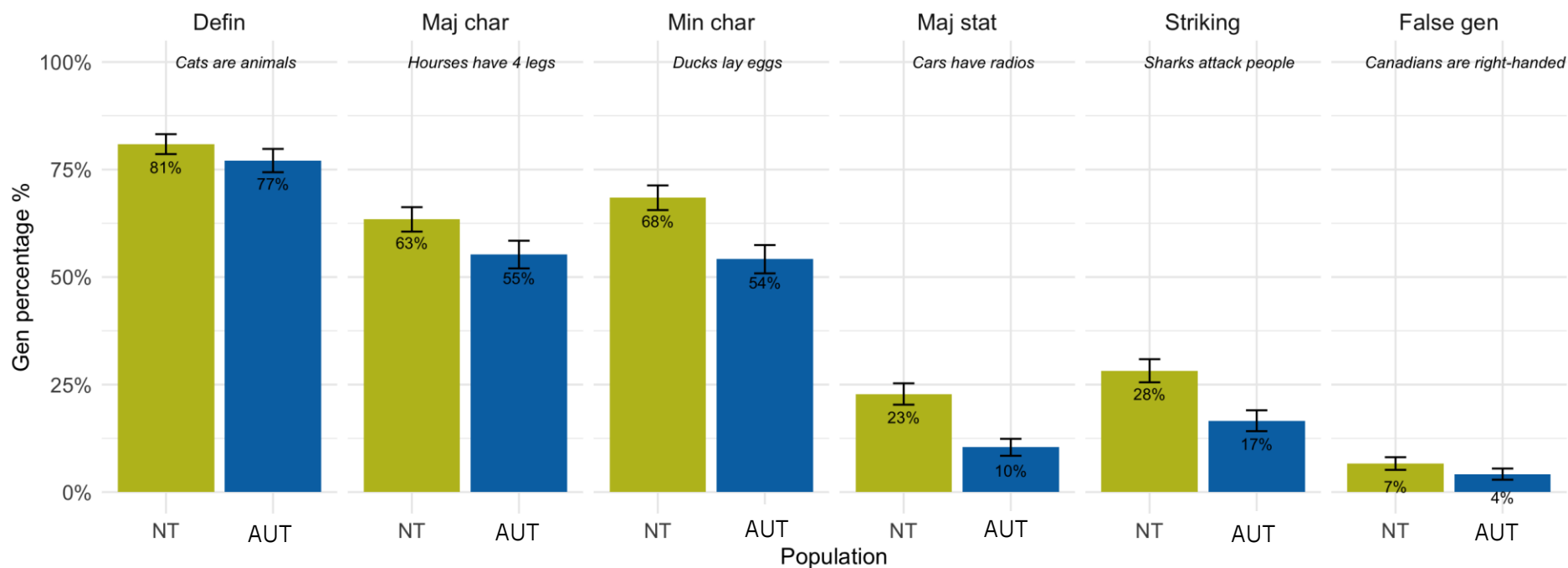
- 48 critical items (8 per statement type)

Statement type	Example
<i>Definitional</i>	Cats are animals
<i>Majority Characteristic</i>	Horses have four legs
<i>Minority Characteristic</i>	Ducks lay eggs
<i>Majority Statistical</i>	Cars have radios
<i>Striking</i>	Sharks attack people
<i>False generalisation</i>	<i>majority:</i> Canadians are right-handed <i>minority:</i> Tigers are albino

- 24 bare plural statements, e.g., *pigs have wings*, which can be made true only by adding *no*
- 24 (bare) mass noun statements, e.g., *fire is hot*, true with a bare singular, which must be left blank

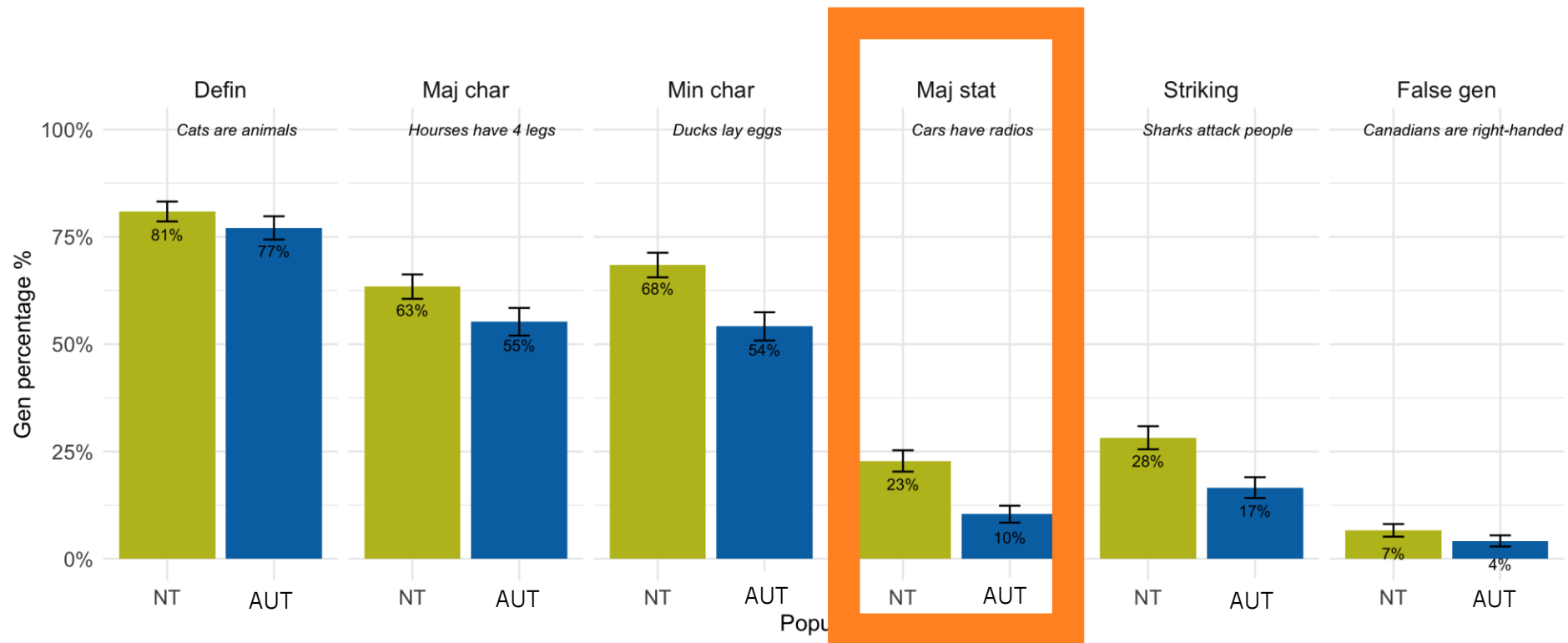


Results: Gen percentage by group



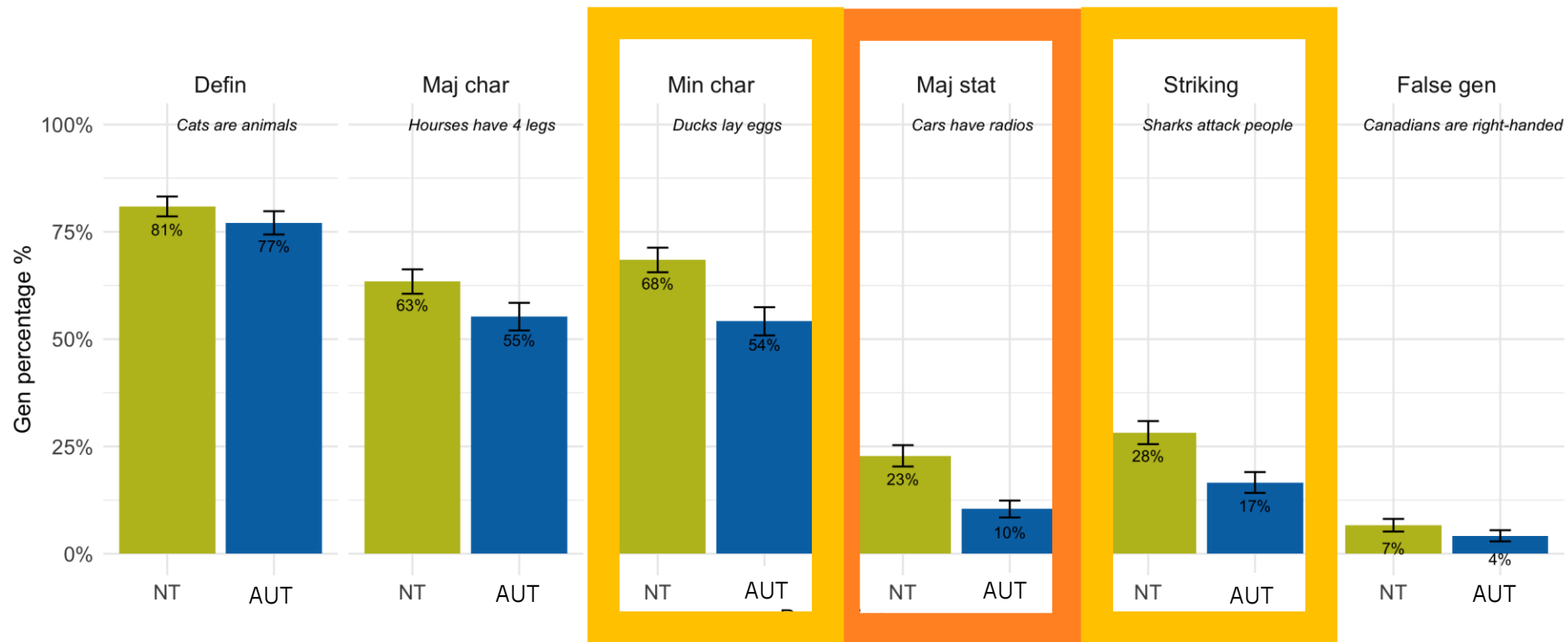
(Group: $\chi^2(1) = 2.630$, $p = 0.105$,
 GroupxStatementType: $\chi^2(1) = 6.502$, $p = 0.260$)

Results: Gen percentage by group



($\theta = -1.222$, $p < 0.05$)

Results: Gen percentage by group



(Striking: $\beta = -1.002$, $p = 0.064$,
Min characteristic: $\beta = -1.005$, $p = 0.056$)

Discussion

- Trends in line with the Nuanced Precisification Hypothesis for the production of generics in autism (min char and striking statements)
- Adults with autism seem more reluctant to produce striking generics than neurotypical adults
 - Do they rely more on prevalence?
- Asymmetry between comprehension and production
- NT vs. AUT category didn't predict responses
 - Ongoing recruitment will focus on individuals with an autism diagnosis

Conclusions

- Striking generics are interesting
- The acquisition studies showed that 4-to-5 year old children are sensitive to strikingness and are developing towards an understanding of its interplay with exception tolerance
- Individuals with autism behave similarly to neurotypicals in the comprehension of generics, but we see differences in production in line with the hypothesis proposed especially in the case of striking generics

THE END THANK YOU!



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